# The Power of Noise: Redefining Retrieval for RAG Systems

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# Motivation: When LLMs Know Too Little

**Large Language Models are impressive — but limited:**

- Knowledge is **frozen** at training time
- Prone to **hallucinations** on factual questions
- Struggle with **domain-specific** or rapidly changing knowledge

Asking a 2023 LLM about last week's events → unreliable answers

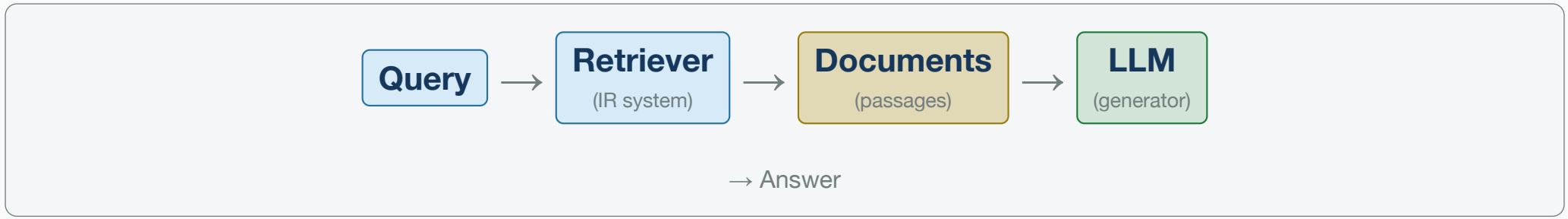
**Retrieval-Augmented Generation (RAG) as the fix:**

- Augment the prompt with **retrieved passages** at inference time
- Grounds the LLM in real, up-to-date evidence
- Standard in enterprise & search applications (Bing, ChatGPT, etc.)

**But:** the retriever's role itself has received **surprisingly little** systematic study.

→ This paper asks: **what should a RAG retriever actually retrieve?**

# How RAG Works — and What We Don't Know




**Common assumption:**

- Feed top-$k$ highest-scoring retrieved docs to the LLM
- More semantically relevant = better
- Number of docs limited only by context window

**Contributions:** first comprehensive study of doc type in RAG; new retrieval heuristics; all code & data released.

**This paper's research question:**

> *"What characteristics are desirable in a retriever to optimize prompt construction for RAG systems in order to increase the LLM effectiveness?"*
>
> Three dimensions studied: **type**, **position**, and **number** of documents.

# A Taxonomy of Retrieved Documents

The paper formalises four document types a retriever can return:

★ Gold — the **ground-truth passage** containing the answer.

● Relevant — other passages that **also contain the answer** (not directly tested).

▲ Distracting — semantically close to the query but **no answer** (top-$k$ but wrong).

■ Random/Noise — **unrelated** to the query; pure noise.

**Prompt structure notation:**

| | |
|---|---|
| **I** | Task instruction |
| ★ | Gold document |
| ● | Relevant documents |
| ▲ | Distracting documents |
| ■ | Random documents |
| **Q** | The query |

Example: `[I, ▲, ★, Q]` means: instruction, then distracting docs, then gold, then query.

The **position** of each document relative to Q will matter greatly.

# Experimental Setup

### 📄 Dataset

**NQ-open** (Natural Questions)

- 21M Wikipedia passages (100 words each)
- 2,889 test queries
- Answers ≤ 5 tokens
- Open-domain Q&A

### 🤖 Generators

- **Llama2-7b** (main model)
- Phi-2
- Falcon-7b
- Mosaic MPT-7b

All open-source, 4-bit quantised
Greedy decoding, max 15 tokens

### 🔍 Retrievers

- **Contriever** (dense, bi-encoder)
- **BM25** (sparse, term-based)

Dense vs. sparse comparison

**Metric:** Accuracy — at least one reference answer must appear in the generated response

**Significance:** Wilcoxon signed-rank test ($p < 0.01$) throughout

# Finding 1: Position of the Gold Document Matters

Three positions tested for the gold document ⋆ **Gold** relative to the query **Q**:

**Near:** `[I, ▲, ⋆, Q]`
Gold adjacent to query

**Far:** `[I, ⋆, ▲, Q]`
Gold at start of prompt

**Mid:** `[I, ▲, ⋆, ▲, Q]`
Gold buried in the middle

**Near > Far > Mid.** The middle of the context is hardest to attend to — worse than the start of the prompt.

**Intuition from attention patterns:**

- Attention scores are **highest near the query** — and also relatively high at the very start of the prompt
- The **middle** of the context gets the least attention: a U-shaped curve
- This is exactly the **"Lost in the Middle"** effect (Liu et al., 2023)

**Practical implication:** place relevant docs **immediately before** the query. Start of prompt is second best — never bury them in the middle.

Consistent across all tested models (Llama2, Phi-2, Falcon, MPT).

# Finding 2: Top-Ranked Docs That Miss the Answer *Hurt*

### Standard assumption in RAG:

- Retrieve top-$k$ documents by similarity score
- Semantically close → helpful context

### What the paper shows instead:

Distracting documents ▲ **Distracting** (high retrieval score, **no answer**) **decrease** LLM accuracy vs.
using only the gold document.

### Example from the paper:

Query: *"What color was Napoléon's horse?"*

A doc about Joséphine de Beauharnais' horse scores high in the retriever (same historical context), but **does not contain the answer**.

### Baseline comparison (Llama2, Contriever, Near):

| | | |
|---|---|---|
| Only gold | `[I, ⋆, Q]` | **0.564** |
| Gold + 1 distracting | `[I, ▲, ⋆, Q]` | ↓ **0.428** |
| Gold + 2 distracting | `[I, ▲▲, ⋆, Q]` | ↓↓ **0.397** |

The **retriever's ranking score is not a reliable signal** for downstream LLM usefulness. High BM25/dense score ≠ helpful in the prompt.

The paper's attention analysis shows the LLM **disproportionately attends** to distracting docs over the gold doc, explaining the wrong answers.

Confirmed with ADORE (hard-negative retriever) — same degradation.

# Finding 3: Random Noise *Improves* Accuracy — Oracle Setting

**Oracle setting:** the gold document is always included. Random Wikipedia passages are added around it.

Prompt: `[I, ▇▇▇, ⋆, Q]` — noise + gold near Q

**Results (MPT-7b, Near):**

| | |
|---|---|
| Only gold `[I, ⋆, Q]` | 0.215 |
| 1. 4 random docs (best) | **0.293** |

**+36% accuracy improvement** from adding documents that contain no answer at all!

**Not consistent across all models:**

- **MPT:** noise helps in *all* positions (Near, Far, Mid)
- **Llama2 & Phi-2:** noise helps only in the **Near** setting; hurts in Far/Mid
- **Falcon:** noise does *not* improve accuracy in this oracle setting

The benefit of noise is **model-dependent** in the oracle setting — but Near positioning + noise is the most robust combination.

**Best noise position:** far from query; gold near query. `[I, ▇▇▇, ⋆, Q]` outperforms `[I, ⋆, ▇▇▇, Q]`.

# Finding 4: Noise Helps in a Realistic RAG Setting Too

**Realistic setting:** retrieve with the IR system, then pad remaining context with random noise.

Prompt: `[I, ■ ■ ■, ▲/●, Q]`

**Accuracy (Llama2-7b, NQ-open test set):**

| | |
|---|---|
| 4 retrieved only (Contriever) | 0.187 |
| 1. fill with noise (Contriever) | **0.253** (+35%) |
| 4 retrieved only (BM25) | 0.203 |
| 1. fill with noise (BM25) | **0.293** (+44%) |

Noise boost **carries over** to the realistic setting — and holds for different noise sources (Wikipedia, Reddit, random words).

**The retriever trade-off:**

| | |
|---|---|
| **Few retrieved (3–5)** | mostly relevant → high signal |
| **Many retrieved (8+)** | distractors creep in → accuracy drops |

**Falcon:** did *not* improve with noise in the oracle setting, but **does improve** in this realistic setting — suggesting the benefit is more general than the oracle results imply.

**Sweet spot:** retrieve **3–5 docs**, pad context with noise. More docs → more distractors.

# Why Does Noise Help? The Attention Entropy Hypothesis

**Attention entropy experiment:**

Compare `[I, ⋆, Q]` vs `[I, ■■, ⋆, Q]`:

Attention entropy **increases ≈ 3×** when random documents are added.

**What is "entropy collapse"?**

Prior work (Attanasio et al., 2022; Hoffmann et al., 2023) shows that:

- Pathologically **low** attention entropy → degenerate, repetitive outputs
- Also linked to worse factual grounding

**Hypothesis:** random documents act as "attention anchors," preventing the model from collapsing onto a narrow region of the context. Higher entropy = more distributed, more robust attention → better answers.

This is an **open hypothesis**, not a proven mechanism. The paper shows **correlation** (noise → higher entropy → better accuracy) but not causation. Explicitly flagged as future work.

# Practical Heuristics from the Paper

### Proposed retrieval strategy:

- Retrieve **3–5 documents** with your IR system
- Place them **near the query** in the prompt
- Fill the remaining context with **random documents**
- Put random filler **far from the query**

Result: `[I, ■ ■ ■, ▲/●, Q]`

### Why not retrieve more?

More retrieved docs → increasing chance of **distracting** content → accuracy degrades.

The sweet spot for the studied setup: $k = 3$ **to** $5$.

### Key insight for the field:

Retrieval quality for RAG is **not** the same as classic IR relevance ranking. A retriever optimised purely for top-$k$ relevance may **hurt** the downstream LLM. New retrieval objectives are needed.

# Connecting to the Course

**Course concepts in this paper:**

**RAG** (Week 4)
The RAG lecture introduced this exact problem: semi-relevant docs as distractors and Lost in the Middle. This paper provides the empirical deep-dive.

**Dense retrieval — Contriever** (Week 3)
Bi-encoder from the Dense Retrieval lecture; used as the main dense retriever in the paper.

**BM25** (Weeks 1 & 3)
TF-IDF weighting (Week 1) → BM25 as sparse retriever (LSR lecture). Paper compares sparse vs. dense for RAG.

**Recall@$k$** (Week 2)
Standard metric from the Evaluation lecture — but the paper shows high Recall@$k$ does **not** guarantee good LLM accuracy.

**What this paper adds beyond the course:**

> Classic IR optimises for **document relevance to query**. RAG requires optimising for **LLM generation quality given the prompt**. These are **not the same objective**.

The Week 4 RAG lecture flags distractors and Lost in the Middle as open challenges. This paper **quantifies** both — and reveals the surprising noise effect not discussed in the course.

The paper's call for new retrieval objectives aligns with the course's progression: classical ranking (Weeks 1–2) → neural retrieval (Week 3) → generation-aware retrieval (Week 4).

# Conclusions

**Three key findings — remember these:**

- **Position matters.** Place relevant docs **near the query** — the LLM cannot attend well to distant context.
- **Distracting docs hurt.** Non-relevant top-$k$ docs **reduce** accuracy.
- **Noise helps.** Random docs **improve** accuracy by up to 35–44%.

**Published at SIGIR 2024.** Code & data released at
`github.com/florin-git/The-Power-of-Noise`
129 citations as of Feb 2026.

**Open questions & future work:**

- **Why** does noise help? The entropy hypothesis is suggestive but not proven.
- Which properties make a document useful as filler?
- Can we **train** a retriever with an objective aligned to LLM accuracy, not just relevance?
- Do these findings hold for larger LLMs (13B, 70B)?
- How do results change with instruction-tuned / RLHF models?